

We would like to begin by thanking the External Evaluation Commission (EAC) for its Preliminary Report, whose observations and conclusions have given us the opportunity, on the one hand, to point out the points for improvement in the educational project that has been submitted for accreditation, so that the necessary measures can be taken to ensure its proper pedagogical and scientific functioning; on the other hand, to clarify some issues that in the CAE's assessment of the UX/UI Design degree programme (Process NCE/24/2400281), the proposing institution made less clear in the initial report and which did not allow it to fully demonstrate the quality that is intended to be imprinted on this project.

In its report, the institution proposed submitting a request for a new SC, in line with the assumptions of the institution's mission and in particular with a strategy for the evolution of teaching, research and know-how, as it is currently doing in its training offer in the areas of Engineering and Computer Science that it already offers, in which technology is of particular importance, given the challenge and the need to be able to respond to the resurgence of industrial activities in Portugal, which require a diversified offer in technological areas, along with the most varied areas of business sciences that the institution also offers, with the aim of contributing to greater competitiveness of the national economy in international markets.

The CAE has assessed the proposal and suggests that it be accredited for a period of 6 (six) years, presenting the justifications for this recommendation, as well as a set of conditions for improvement in the proposal submitted.

We are therefore grateful for the trust that has been placed in us, as well as for the work carried out by the CAE, to which we give deserved value and which we will naturally take due account of.

Also according to CAE, the SC suffers from a certain lack of identity, sometimes appearing to be more of a cluster of topics around UX/UI than a well-planned training strategy for its students.

It's important to note here that the SC identifies with the assumption that the evolution of Engineering and Technology, with the contribution of Design, has progressed along two fundamental axes: (1) increasing complexity and (2) ease of use.

In this respect, we know that today we have at our disposal a variety of solutions that are immeasurably more complex and that are very easy to use in our daily activities.

Today, user experience and interface design are highly valued, from the screens we touch and run our fingers across to the services we rely on. We therefore understand that UX and UI Design is an emerging field of multidisciplinary skills and activities that design future-orientated platforms, including: applications, websites, digital products and services, augmented and virtual reality experiences, installation design, wearable technology and the Internet of Things.

This is effectively the purpose of this 1st cycle course in UI/UX Design: it is supposed to be practice-orientated, and its structure over its six semesters includes 360 contact hours of practical-laboratory work (48.06% of the total contact hours of the compulsory courses). With this in mind, each semester includes one or more CUs with practical skills and project practice, with special emphasis on the final semester, in which there is a CU called 'Final UX/UI Design Project', with a total of 180 contact hours, distributed according to the following types: PL - 105; TP - 45; S - 15; OT - 15, corresponding to 18 ECTS.

In addition to the practical CUs, there are other CUs that analyse the fundamentals, including user studies, experience design, interaction design, visual design and information architecture - and how these different components can be identified, researched, informed and questioned

in the context of professional activity, in line with the principles of Atlântica's pedagogical model.

It also uses 'design thinking' methodologies, human-centred design and the creative and innovative use of technology to drive organisations forward and inspire positive change.

methodologies, in which students are responsible for developing their own skills, competences and specific their own skills, competences and specific knowledge, relying naturally with the support of the teacher as their mentor and inspirer, so that they can fulfil their potential, just as the market is doing today, particularly in the case of start-ups.

The aim is to combine theory with practice in order to understand user behaviour and design products and services that optimise the user experience.

This CE in UX/UI Design thus aims to explore the importance of design for the end user experience with intuitive user interfaces. Students will have the opportunity to develop projects based on user-centred research and design experimentation, which will allow them to showcase their creative, analytical and technical knowledge acquired in the meantime and highly valued in today's technology and business market.

We are convinced that these future UX/UI designers are currently in huge demand in all sectors of activity, thanks to our unique links with industry. As a strategy, this SC is designed to prepare students, equipping them with an appropriate leadership profile, throughout the product development process for the continuous improvement of the user experience in products and services.

As a strategy for emerging training in the area of Design, it aims to provide its future graduates with the aggregation of multidisciplinary skills, providing them with the appropriate tools to develop their careers in one of the most promising areas of today, the digital and technological industry, integrated into a unique ecosystem for innovation and development - Oeiras Valley - as a hotbed of innovation, creativity and technology in Portugal, which favours the installation of start-ups or technology-based companies.

Review of UC programmes, their syllabus, learning outcomes, teaching methodologies and assessment; Review of overlapping content between UCs; Review of articulation between UCs in the same semester and between semesters

Now that we are analysing the details with a view to correcting or implementing the recommendations and considerations listed by the CAE in its report, point 4.6.1. 'Overall assessment', we have made a significant number of adjustments with the coordinating team and the respective teachers responsible for the courses.

However, it will still be possible in the future to analyse their articulation and possible adjustments during the course of the SC's eventual operation, with the aim of optimising its pedagogical organisation, clarifying the dynamics and hierarchies of the various courses and their relationships, as well as revising the programmes of the courses and their syllabuses, learning outcomes, teaching methodologies and assessment, as well as reviewing overlapping content between courses, updating or reviewing the bibliography, as well as reviewing the links between courses in the same semester and between semesters, we will now detail the changes made step by step, as well as the clarifications that are necessary:

CU: Algorithms and Programming

(The programme mentions web technologies that are not included in the content; The main language used seems to be Python, but the references do not include resources on Python; The assessment dedicates 60% of the grade to a critical essay, which does not seem appropriate for an introductory

programming course; This course is only offered in the fourth semester, so it might make sense to teach it earlier).

With the technological evolution underway; given that web technologies are by nature very dynamic, it is to be expected that over the course of the course, it may include new technologies to be integrated into the syllabus to be presented and developed in this course.

In any case, we have opted to include a reference associated with Python and propose updating the list of bibliographical references:

- 1/ Bohnacker, H.; Groß, B.; Laub J.; Lazzeroni, C. (2010). Generative Design: Visualise, Program, and Create with Processing: Princeton Architectural Press.
- 2/ Groner, L. (2020). *Python for designers: Learn Python to automate design workflows*. Independently Published.
- 3/ Burch, M. (2022). *Creating graphical user interfaces with Python: Using Tkinter and Qt*. Apress.
- 4/ Cormen, Leiserson, Rivest, and Stein (2009). Introduction to Algorithms, 3rd Edition, MIT Press.
- 5/ Paul McFedries (2023). HTML, CSS, & JavaScript All-in-One For Dummies.

As for the assessment criteria, this is understood as an oversight; it will be rectified by creating an algorithm for a topic to be worked on in line with market trends at the time the course is taught.

As for the UC's position in the SC's curricular structure, the observation is pertinent; it would indeed make sense for it to be integrated into the previous semester. However, the target audience of students that we want to attract is probably mostly from the arts and humanities. They may not be prepared for a course of this nature at an earlier stage in the programme. For this reason, it was decided to better prepare students in more technological practices and then involve them in activities that require content focused on the mathematical sciences, as is the case with this CU.

CU: Creativity

(Although it is understandable to offer a CU dedicated to design as a creative practice in the first semester, the objectives and methodologies listed are unclear, a problem that begins with the very definition of 'creativity'; it is unclear how the programme will be implemented, whether through classroom exercises or projects).

In view of the considerations raised, we have taken the initiative to reformulate the learning objectives for this CU:

- C2 - Identify works by reference authors and understand the concepts of creativity, innovation and their role in UX/UI Design;
- C1 - Identify creative tools and techniques for creating aesthetically appealing and functional interfaces;
- C3 - Identify and explain project development methodologies and methods in the field of design and in particular production with digital technology;
- Ap1 - Critically reflect on the importance of creative performance, creativity and innovation;
- Ap2 - Develop creative and innovative thinking skills applied to UX/UI Design;
- At2 - Apply creative techniques in collaborative processes, such as brainstorming, mind maps, SCAMPER and storyboarding in the design process.

In line with the learning objectives to be achieved, the following reformulation of the demonstration of the consistency of the syllabus with the learning objectives of the course is also proposed:

Learning objective C1 will be worked on in CP1 and CP8 of the syllabus addressing concepts of creativity from different authors.

Objective C2 will be developed in CP2 to CP5 where the different dimensions, characteristics, facilitating and inhibiting factors and processes of creativity will be addressed.

Objectives Ap1, Ap2 and At2 will be worked on in CP6 to CP9 by adopting creative methodologies and ways of promoting innovation through the creation of practical exercises in the classroom that encourage experimentation and interaction.

CU: Applied Design

(The programme covers various types of design in a one-semester course, which raises questions about the learning outcomes for students, who will have very limited time to devote to each type; the bibliography focuses exclusively on computer-aided design, ignoring at least two-thirds of the programme).

In view of the considerations listed above, we believe it is best to adapt the syllabus to enable students to develop drawing skills applied to the digital environment, using tools, techniques and methodologies that support design in areas such as illustration, graphic design, animation and digital interfaces:

- CP1: Introduction to Digital Design: fundamental concepts of design in the digital environment.
 - CP1.1: The difference between traditional drawing and digital drawing.
 - CP1.2: The importance of digital drawing in different areas: illustration, graphic design, UX/UI, animation and games.
- CP2: Drawing fundamentals
 - CP2.1: Lines and shapes: practice and precision in digital drawing.
 - CP2.2: Perspective and proportion:
 - CP2.2.1: Drawing in 1, 2 and 3 vanishing points.
 - CP2.2.3: Construction of three-dimensional objects and scenery.
- CP3: Drawing Characters and Visual Elements
 - CP3.1: Building characters:
 - CP3.1.1: Basic anatomy (human proportions and stylisation).
 - CP3.1.2: Facial expressions and dynamic poses.
 - CP3.1.3: Creating clothes and accessories.
- CP4: Drawing graphic elements: icons, objects and symbols.
- CP5: Vector Drawing and Illustration
 - CP5.1: Difference between raster art and vector art.
 - CP5.2: Creating vector illustrations in Adobe Illustrator.
 - CP5.3: Pen tools and basic shapes.
 - CP5.4: Use of Bézier curves and stroke precision.
 - CP5.5: Application to logos, icons and UI/UX elements.
- CP6: AI as a design tool.
 - CP6.1: Building prompts.
 - CP6.2: Analysing and using computer images.

The compulsory bibliography will be updated to:

- 1/ Loomis, A. (2011). Figure Drawing for All It's Worth. Edition: Titan Books.
- 2/ Edwards, B. (2012). Drawing with the Right Side of the Brain. Ediouro Edition.
- 3/ Ruiz, M. (2016). Digital Painting Techniques: Practical Techniques of Digital Art Masters. Edition 3DTotal Publishing.
- 4/ Bancroft, T. (2016). Creating Characters with Personality. Watson-Guption Edition.

5/ Robertson, S. (2013). How to Draw: Drawing and Sketching Objects and Environments from Your Imagination. Design Studio Press edition.

UC: Digital Product Development

(The methodology and assessment are unclear; The learning objectives highlight cloud computing, but this is not mentioned in the programme; The contact hours appear to be very limited in relation to the objectives listed).

It is important to clarify that, as an optional course of a theoretical-practical nature, the contents and objectives listed are in line with Atlântica's pedagogical model, using active methodologies, where students are encouraged to build their own knowledge and turn to teachers as mentors of that knowledge. In this respect, we believe that the contact hours of both this and the other CUs on this course fulfil the most up-to-date expectations associated with teaching-learning methodologies and adopted by most of the best higher education institutions.

With regard to the aforementioned omission, the syllabus has been corrected:

- CP1: Introduction to Digital Product Development
- CP2: Digital Transformations
- S3: Basic concepts of Cloud Computing
- CP4: Methodology for Product Design
- CP5: Agile Project Management
- CP6: Digital Solutions Architecture
- CP7: Product Growth
- CP8: People and Team Management
- CP9: Product Design
- CP9.1: Preconception
- CP9.2: Conception
- CP9.3 Post-conception

UC: ESG, Design and Marketing

(The learning outcomes are generic and appear to overlap with those of the Entrepreneurship UC; the bibliography does not correspond to the programme).

In view of the considerations raised, we have taken the initiative to reformulate your learning objectives:

- C1 - Identify and understand ESG principles and their strategic application in design and marketing;
- C3 - Identify and explain methodologies and methods of project development in the area of sustainability and in particular production with digital technology;
- Ap3 - Evaluate the environmental and social impact of marketing and design strategies;
- At1 - Develop critical capacity to evaluate corporate governance practices and their communication to the public;
- At2 - Develop entrepreneurial and creative skills, proactivity, leadership of multidisciplinary teams and intervention, in line with the learning objectives for sustainable development and ESG practices;
- At3 - Create visual and functional solutions that communicate ESG values clearly and effectively.

The compulsory bibliography will also be updated:

- 1/ Papanek, V. (2005). Design for the real world: Human ecology and social change (2nd ed.) Thames & Hudson.

- 2/ Kotler, P., & Keller, K. L. (2022). Marketing management (16th ed.). Pearson.
- 3/ Carroll, A. B., & Buchholtz, A. K. (2014). Business & society: Ethics, sustainability, and stakeholder management (9th ed.). Cengage Learning.
- 4/ Manzini, E. (2015). Design, when everybody designs: An introduction to design for social innovation. MIT Press.
- 5/ The-Sustainable-Development-Goals-Report-2022. United Nations (available at <https://unstats.un.org/sdgs/report/2022/The-Sustainable-Development-Goals-Report-2022.pdf>);

UC: Aesthetics and Design

(Unclear learning outcomes; The bibliography does not cover the fields of computational and interaction design; The methodologies do not seem adequate).

In order to make the learning objectives clearer, we propose the following change:

- C1 - Identify national and international reference works and authors in the field of Aesthetics;
- C2 - Deepen the ability to critically analyse contemporary design production according to aesthetic reasoning and language.
- C3 - Understand the fundamental concepts of aesthetics applied to UX/UI Design.
- Ap1 - Critically analyse aesthetic choices in different UX/UI Design contexts in professional practice.
- Ap2 - Apply design aesthetics to develop UX/UI Design digital artefacts.
- At1 - Develop practical skills in creating designs with solid aesthetic foundations.
- At3 - Explore the relationship between aesthetics, functionality and the user experience.

As far as methodologies are concerned, we believe that they are in line with Atlântica's pedagogical model and in particular with the other predominantly theoretical CUs in the programme.

The compulsory bibliography will be updated to:

- 1/ Flusser, V. (2007). The Universe of Technical Images: Praise of Superficiality. São Paulo: Annablume.
- 2/ Cardoso, R. (2004). Design for a Complex World. São Paulo: Cosac Naify.
- 3/ Bense, M. (1965). Generative Aesthetics. Stuttgart: Edition Rot.
- 4/ Barthes, Roland (2007). Criticism and Truth, Lisbon, Edições 70.
- 5/ Berger, John (2018). Ways of Seeing, Lisbon, Antígona.

UC: Design Management

(Some elements of the programme seem to overlap with other UCs without it being clear which articulation strategies are to be developed; The programme mixes marketing tools and methodologies with design project tools and methodologies (e.g. personas, storytelling, etc.).

It is our understanding that this CU addresses issues and concepts that we consider essential for strategic design and the efficient management of UX/UI processes, promoting the training of professionals capable of leading teams, as well as aligning business objectives with the eventual marketing teams and the actual needs of the end user.

UC: History and Theories of Design

(Four of the five references listed are more than thirty years old; Bonsiepe's name is misspelt; Bürdek's book had a second edition updated in 2005).

As this is a History of Design course, we initially opted for a bibliography that we considered to be classic. However, we updated it in line with the other observations made by the CAE:

- 1/ MARGOLIN, Victor.(2017). World History of Design. New York: Bloomsbury Publishing
- 2/ RAIZMAN, David (2023). History of Modern Design (Third Edition). London: Lawrence King. 448p.
- 3/ BÚRDEK, B., (2005). Diseño, história, teoria y práctica del diseño industrial (second edition), Barcelona, Gustavo Gili. 390P.
- 4/ BONSIEPE, G., (1992). Teoria e prática do design industrial, Lisbon, CPD. 362p.
- 5/ MILLER, J. Abbott, LUPTON, Ellen.(2019) The Abc's of Triangle Square Circle: The Bauhaus and Design Theory. Princenton: Princeton Architectural Press. 72p.

UC: Digital Illustration

(There seems to be a significant overlap in content between this UC and Applied Design; the bibliography is quite technical, with few historical and theoretical references).

We believe that this is precisely the purpose of these two CUs mentioned by the CAE. In the specific case of this Digital Illustration CU, it aims to focus on the application of drawing much more effectively and using digital technologies for applications of this nature.

As for the bibliography being quite technical, with few historical and theoretical references, as we mentioned earlier, this is indeed its purpose, given that this CU is a sequel to the Applied Drawing CU, where historical and theoretical references are covered. This CU essentially aims to strengthen the practice of Applied Design using the most appropriate digital tools at the time of its application.

UC: Artificial Intelligence Applied to Design

(The programme seems to be more comprehensive than that of the Introduction to Artificial Intelligence UC, which raises questions about the need for both UCs or their linkage).

This observation on the part of the CAE is, in fact, very relevant for the purpose of dividing the content and knowledge acquired within the scope of 'Artificial Intelligence' into two CUs. While the first UC, 'Introduction to Artificial Intelligence', was more theoretical and comprehensive in nature, this UC, 'Artificial Intelligence Applied to Design', essentially focuses on the application of 'Artificial Intelligence' in design practices. It should be understood that the target audience for this course is expected to come mostly from the arts and humanities. In this respect, it is our understanding that these students need to be better prepared so that they can better use and adopt the use of digital technologies using maths, such as algorithms and artificial intelligence. In fact, it is crucial for this teaching-learning strategy to keep these two CUs in the CE structure, so that better results can be achieved and the desired success can be achieved for the students.

CU: Lab Interface Design

(The name of the course does not correspond to the programme, which is mainly dedicated to typography).

In view of the changes that have since been proposed to the syllabus of the Lab Visual Design course, which precedes this course, we propose to amend and reformulate the following sections of the FUC:

Learning objectives and their compatibility with the teaching method (knowledge, skills and competences to be developed by students):

- C1 - Understand the importance and application of visual design principles to create clear, aesthetic and functional interfaces;
- C2- Understand the importance of appropriate typography and layout for interfaces.
- Ap1 - Apply interface design principles to create user-centred solutions;.
- At1 - Develop interfaces with a focus on usability, accessibility and aesthetics;
- At2 - Be able to carry out usability tests and implement iterative improvements.

Programme contents:

- CP1: Visual Interface Design
 - CP1.1: History and technological advances in typography: analogue and digital.
 - CP1.2: Typography applied to interface design:
 - CP1.3: Selection of typography:
 - CP1.3.1: Criteria for digital interfaces.
 - CP1.3.2: Typographic hierarchy: headings, subheadings, body text and subtitles.
 - CP1.3.3: Legibility and accessibility: text sizes, weights and alignment.
 - CP1.3.4: Use of fonts on different platforms (web-safe fonts, Google Fonts).
 - CP1.4: Microinteractions and animations: enhancing the user experience.
 - CP1.5: Studying motion design and visual feedback.
- CP2: Accessibility and Inclusive Design
 - CP2.1: Principles of Web Accessibility (WCAG)
 - CP2.2: Interfaces for different devices and needs (e.g. screen readers, contrast adaptation).
 - CP2.3: Inclusive design: diversity of users and contexts.

Demonstration of the consistency of the syllabus with the learning objectives of the course:

- Objectives C1 and C2 will be worked on in CP1 addressing, in particular, the principles and theories underlying typography and layout in a conscious and judicious manner.
- Objective Ap1 will be worked on in CP2 where product analysis and practical interface design exercises will be carried out to consolidate the content.
- At1 and At2 will be worked on in CP2 and throughout the course through usability, accessibility and aesthetic evaluation activities that promote iterative improvements.

Bibliography to be consulted/mandatory:

- 1/ Lidwell, W., Holden, K., & Butler, J.(2010). Universal Principles of Design. Rockport Publishers.
- 2/ Norman, D. A. (2013). The Design of Everyday Things: Revised and Expanded Edition. Basic Books.
- 3/ Garrett, J. J. (2010). The Elements of User Experience: User-Centered Design for the Web and Beyond. New Riders.
- 4/ Shneiderman, B., & Plaisant, C. (2010). Designing the User Interface: Strategies for Effective Human-Computer Interaction. Pearson.
- 5/ Cooper, A., Reimann, R., & Cronin, D.(2007). About Face: The Essentials of Interaction Design. Wiley.

UC: Lab Design Visual

(The programme seems to correspond more to an introduction to graphic design than to a UC on visual design).

Our aim would be an introduction to Graphic Design with a focus on Visual Design, but in view of the above considerations, we propose changing the syllabus to:

- CP1: Introduction to Visual Design and User Experience (UX)
 - CP1.1: Fundamental concepts: what is UX Design and the role of Visual Design.
 - CP1.2: The relationship between functionality, usability and aesthetics.

- CP1.3: The importance of visual design in the user experience.
- CP1.4: Examples of good and bad visual design practices in digital interfaces.
- CP2: Principles of Visual Design
 - CP2.1: Gestalt: perception of shapes, formal analogies and their application in UX: proximity, similarity, continuity, figure-ground, closure.
 - CP2.2: Visual hierarchy: use of size, contrast, colour and spacing.
 - CP2.3: Principles of alignment and grids for visual organisation.
 - CP 2.4: Applied typography: legibility, scalability and typographic harmony.
 - CP2.5: Choosing and combining fonts.
 - CP2.6: Colour palettes: colour psychology and its impact on the user experience. Colour accessibility.
- CP3: Visual Design Tools
 - CP3.1: Practical exploration of the main tools on the market:
 - CP3.1.1: Figma: design, prototyping and collaboration.
 - CP3.1.2: Adobe XD: wireframes, interactive prototypes and testing.
 - CP3.1.3: Sketch: vector interface design.
 - CP3.2: Complementary tools:
 - CP3.2.1: Colour palettes: Colors, Adobe Color.
 - CP3.2.2: Iconography: Material Icons, Font Awesome.

CU: Video Lab

(The UC's English name is 'Video Lab', in line with the 'Lab' of Visual Design and Interface Design, while the Portuguese name is 'Laboratório'; The most recent source in the bibliography is almost 10 years old, some of the others are more than 20 years old. In a very technical course, the references should reflect the state of the art).

Following the recommendation of the CAE and in agreement with other CUs in the curricular structure that are similar, we propose adopting the Portuguese name for the **CU: Lab Vídeo**.

The compulsory bibliography will be updated to:

- 1/ Zettl, H. (2017). Television and video production: Techniques, aesthetics and practices (10th ed.). Pioneer Thomson Learning.
- 2/ Thompson, R. (2018). Digital video production handbook [eBook]. Elsevier.
- 3/ Costa, M., & Lopes, F. (2021). Aesthetic analysis in the production of short films in a video laboratory. *Revista de Artes Visuais*, 15(3), 45-60.
- 4/ Pereira, A. C. (2019). The use of editing techniques in experimental audiovisual narrative. *Comunicação e Cinema*, 8(2), 20-34.
- 5/ Marques, P. (2020). Lighting and visual composition in video. In J. Silva (Ed.), *Manual prático de vídeo digital* (pp. 101-130). Edições Lusíadas.

UC: Digital Marketing

(UC Fundamentals of Marketing already includes an introduction to Digital Marketing; SEO and Social Network Management also includes content on Digital Marketing).

Given the need for a better relationship and articulation between the above-mentioned CUs and to make the course more dynamic, including allowing students who opt for the Marketing Fundamentals CU to have access in advance to content of an introductory nature to Digital Marketing or complementary in the case of the SEO and Social Network Management CU, we believe it is relevant to maintain them as proposed.

CU: Visual Merchandising

(Unclear learning objectives and potentially redundant with those of other CUs).

We have updated the learning objectives:

C2 - Understand the importance of the techniques used by retailers in merchandising and displaying merchandise in real retail digital environments.

C3 - Understand the different approaches and methods for effectiveness in real retail digital environments, taking into account display principles of balance, colour and focal point assertion.

Ap1 - Apply visual merchandising principles to e-commerce platforms and apps.

Ap2 - Create intuitive and engaging shopping experiences in line with consumer needs.

Ap3 - Develop information architecture and efficient navigation flows for digital merchandising.

At1 - Implement micro-interactions and animations to improve the shopping experience.

At3 - Evaluate the performance of visual merchandising based on data and conversion metrics.

CU: Design Methodologies

(The name of the CU is different in English (singular) and Portuguese (plural)); The book 'Universal Methods of Design' by Martin and Hanington had a revised and expanded edition in 2019).

The name of the English version of the course is corrected: Design Methodologies.

The reference bibliography has also been updated:

1/ Brown, Tim (2017) Design Thinking. Alta Books; Edition: 1st original version: Change by Design;

2/ Cross (2007) Designerly Ways of Knowing. Basel; Boston; Berlin: Birkhauser;

3/ Jones (1992) Design Methods. New York: John Wiley & Sons, Inc. (1st Ed. 1970);

4/ Kelley (2001) The Art of Innovation. London: Harper Collins Business, 2001;

5/ Martin; Hanington (2019) Universal methods of design: 100 ways to research complex problems, develop innovative ideas, and design effective solutions. Beverly, MA: Rockport Publishers.

In view of the possible overlap of programme contents with other UCs, they will be updated:

CP1: Introduction to the principles and concepts of Design

CP1.1: Simplicity versus complexity

CP1.2: Design methods

CP1.3: Design tools

CP1.4: Design techniques

CP2: Capturing and analysing market data

CP3: Agile methodologies

CP4.1: Concepts

CP4.2: Models

CP4.3: Utilisation

CP4: Co-creation

CP5: Design thinking

CP6: UX metrics and evaluation

UC: Interface Project

(The course is entitled 'Interface project' in Portuguese but 'Interface design' in English; There is a potential overlap with other courses, namely Design Methodologies; Part of the bibliography does not adequately support the content, other references are out of date).

The name of the English version of the course has been rectified: Interface project.

The syllabus for the Design Methodologies course has also been revised.

As for compulsory consultation/existence, it will be updated to:

- 1/ Alan Cooper, Robert Reimann, David Cronin, & Christopher Noessel (2014). About Face: The Essentials of Interaction Design: Vol. Fourth edition Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel. Wiley.
- 2/ Tidwell, J. (2011). Designing Interfaces : Patterns for Effective Interaction Design: Vol. 2nd ed. O'Reilly Media.
- 3/ Nielsen, Jakob, and Loranger, Hoa (2007). Web Usability 2.0: Designing Quality Websites. Editora Alta Books.
- 4/ Garrett, Jesse James (2011). Elements of User Experience: Interactive Design for the Web and Beyond. Bookman Publishing.
- 5/ Krug, Steve (2014). Don't Make Me Think: A Common Sense Approach to Web Usability. Alta Books.

UC: Digital UX/UI Design

(Is the adjective 'digital' necessary?; Some of the learning outcomes are unclear; User research and testing seem to be treated as secondary aspects).

In view of the question raised by the CAE, it is important to clarify that the course focuses on digital design in a broad sense and in the context of technologies. It should also be clarified that, with regard to the points raised about user research and testing, it should be noted that this course focuses more on technologies and not on processes and methods, leaving the stages mentioned for the following project courses and for the end-of-course project.

Given that some of the learning outcomes are unclear in the assessment of the CAE, the following reformulation is presented:

- C2 - Understand the process intrinsic to collecting, researching and utilising data for UX/UI Design.
- C3 - Understand the essential phases of the UX/UI Design cycle.
- Ap1 - Handle modern tools for prototyping and design.
- Ap2 - Enable students to design user-centred solutions.
- Ap3 - Identify the strengths and weaknesses of interfaces and give suggestions on how to improve them.
- At1 - Develop creative and original solutions using prototyping tools.
- At3 - Develop autonomously and know how to defend proposed solutions for digital projects with data, using performance metrics created by the student.

UC: Final UX/UI Design Project

(This course seems to be an expanded version of UX/UI Digital Design, as some references are essentially identical).

We agree with the CAE's assessment. That is indeed the purpose of this CU, as a course 'Final Project'. It is intended to encompass content from the various UCs that preceded it, so that the final project can be realised and, if possible, applied to the market with coherence of content and adopting current methods, tools, techniques and technologies that are available. The methodology followed at Atlântica is structured on the assumption of an active format and this course, in our view, has been structured to bring practical cases associated with the market into the classroom. The realisation of a FINAL PROJECT is understood to be the result of the acquisition of knowledge and skills acquired throughout the course of the students' SC.

This is also the reason, and the matrix, for many of the contents overlapping in some CUs, so that they can build a solid, integrative education.

CU: UI Design

(The programme of this CU is very ambitious, several of the topics seem to need well over a week to be dealt with properly; Some of the topics seem to belong more naturally within the scope of an Interaction Design or Human Factors CU; It is not clear how the CU will be developed as the programme is quite theoretical but the domain of the CU is largely practical).

In view of the assessment made by the CAE, it is important to clarify that the contents and learning objectives listed fit in with the pedagogical model adopted by Atlântica, using active methodologies, where the student should be able to construct their own knowledge and turn to the teachers as mentors of that knowledge, as mentioned above. As such, it is our understanding that the contact hours of both this and the other CUs proposed for this course fulfil the most up-to-date teaching-learning expectations advocated in most of the best higher education institutions.

As for the nature of how the course will be developed, it is important to clarify that the lessons will be supported by real examples and applied exercises, which will encourage critical analysis and discussion of examples of existing interfaces. Individual and group work will also be carried out, with continuous feedback. It is also proposed to explore design tools such as Figma and others, such as Sketch.

CU: UX Design & UX Design II

(The programme for the two courses is very ambitious and not very coherent; There is no clear logic behind the order of the content or even the difference between the two courses; The UX Design II course includes Adobe XD, which is no longer actively developed; There is some overlap of content with other courses, namely UI Design; It is not clear whether the course development strategy will be more practical or theoretical, whether the students will develop a project, whether these courses are project-based or technical).

It should be noted that the mention of Adobe XD and other software mentioned will be removed and replaced with 'Training in software tools in use for UX Design.'

In this regard, the syllabus for **UX Design II** has been updated:

- CP1: Emotional Design
 - CP1.1: Definitions and types of emotion
 - CP1.2: Understanding emotions in UX/UI
- CP2: Levels of Emotional Design
- CP3: User Journey
 - CP3.1: Levels of the journey
 - CP3.2: Storytelling
- CP4: UX Writing
 - CP4.1: Information Architecture
 - CP4.1.1: Data vs. Information
 - CP4.1.2: Taxonomy and Folksonomy
 - CP4.1.3: Uses
 - CP4.2: Brand tone and voice
- CP5: Digital Products
- CP6: Road Map
 - CP6.1: Utilisation
 - CP6.2: Prioritisation
- CP7: Prototyping

CP7.1: Training in software tools in use for UX Design

As far as the methodologies to be adopted within the scope of these courses are concerned, they involve theoretical-practical classes to explain UX concepts and frameworks and case studies: analysing real UX Design projects. In addition to these, practical-laboratory classes are planned, where wireframes, prototypes and usability tests will be developed, supported by group work: developing UX design projects with continuous feedback; and for analysing metrics and tests: interpreting data to improve UX projects.

Review of the scientific areas associated with the UCs

Finally, based on the preliminary report drawn up by the CAE, which mentions the need for a review of the scientific areas associated with the UCs, namely, according to the report, there is an apparent mismatch between their contents and the associated scientific areas, identifying some UCs in this context. We have therefore decided to provide the necessary clarification.

The UCs identified are all optional. These CUs are associated with different option groups and are allocated to different years/ semesters, according to point 4.3 'Curricular Units (options)'. Each option group can offer CUs from the same scientific area or from different scientific areas. It is not at all our intention to create new scientific areas by amalgamating knowledge. This procedure followed the procedures and guidelines provided by the A3ES Guides and in accordance with the fields for submitting proposals of this nature for the optional courses.

Also with regard to the optional courses on offer, the CAE mentions as a weak point that these courses, although available in a given semester, cannot apparently be selected by students outside that semester, which can make it impossible to attend them and seriously limit the individual configuration of the learning path.

We naturally take good note of this observation made by the CAE. We are therefore considering the possibility of giving students the opportunity, regardless of the curricular year in which they are enrolled, to select one of the six optional courses available in that semester, as provided for in the CE's curricular structure of optional courses in each of its three curricular years.

In conclusion, this SC aims to offer a basic and broad training in the area of study, with special emphasis on developing general and introductory knowledge of an interdisciplinary nature that will enable its graduates to enter the labour market, as reflected in the distribution of credits across the different scientific areas, thus giving preference to the introduction of fundamental concepts, theories and practices, along with training centred on practice and know-how, supported by problem-solving, preparing these future graduates to feel qualified to continue their studies - in a possible master's degree course, through more specialised training in the area in question.

Conclusions

This presentation has also shown that:

1. The HEI believes it has corrected and implemented the recommendations and considerations listed by the CAE in its Report in most of the cases highlighted, namely in point 4.6.1. 'Overall assessment', in internal liaison with the coordination of the SC and by the teachers responsible.
2. It also hopes that the terminology used to identify the scientific areas associated with the optional courses has been clarified.